a) If L is the root, we are good and we don't have to do anything.

b) If L has at least one sibling S with more than m points, we can move one point from S to L. Although we would be careful to choose the closest point to L (among all its siblings with at least m+1 points), this operation can potentially cause L's bounding envelope to expand significantly (if only sibling far away from L have enough points) and unbalance the tree.

c) If no sibling of L can "lend" it a point, then we will have to merge L with one of its siblings. Again, we would then have to choose which sibling to merge it to, and we might choose different strategies:

- Choosing the closest sibling;
- Choosing the sibling with larger overlap with L;
- Choosing the sibling that minimizes the coordinates variance (over all axes).

Case 2.c is clearly the hardest to handle; case 2.b, instead, is relatively easy because, luckily, one difference with B-trees is that node's children doesn't have to be sorted, so we don't need to perform rotations when we move one point from S to L. In the middle-bottom sections of figure 10.19 you can see the result of node S3 "borrowing" one of S4 children, S9: just as easy as that. Of course, the hardest part is deciding which sibling to borrow from and which of its children should be moved…